

MLFF Architecture Revision 74

SIZE-FIDELITY2 admission-width survivor requalification

mdstats project

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Revision 74

Revision 74 implements **SIZE-FIDELITY2** as a pre-migration, GPU-deferred scientific control plane over the fixed-eight MV target-size funnel. The authority calibrates every scientifically available admission width $q = 4, 5, 6, 7, 8$ from one exhaustive set of uninterrupted 30-epoch trajectories. Calibration disables halving during execution: each frozen optimizer seed trains every hard-qualified size once to epoch 30, and the 3- and 10-epoch survivor decisions are reconstructed retrospectively from those same checkpoints.

The hard fidelity criterion is finalist recall rather than rank correlation. For each seed and admission width, the complete 30-epoch population defines the two eventual target finalists under the frozen practical-equivalence ordering. Both finalists must survive the reconstructed epoch-3 $q \rightarrow \min(q, 4)$ stage and the epoch-10 $4 \rightarrow 2$ stage. A material advantage of the 16,384 boundary beyond the practical-equivalence width fails the bounded-ceiling authority instead of being interpreted as convergence.

SIZE-FIDELITY2 also inherits the prior monitor-efficiency plan. Monitor sizes 128, 256, 512, 1024 are derived from one authorized epoch-3 full prediction product, so monitor calibration adds zero model inference passes. The smallest monitor giving exact epoch-3 promotion-set equivalence across all seeds and available q widths is recommended.

Campaign **prepare** now persists `size_fidelity2_execution_plan` and includes it in restart receipts. The plan is work-authorizing only when **SIZE-HALVE2** is ready; otherwise it is a fail-closed no-work record. Positive MACE/GPU calibration remains deferred to **FINAL-GPU1**, so this revision does not claim real accelerator survivor recall and does not change revision-64 **TARGET-DATA2C v4**, **TARGET-DATA2D v2**, **DATA8** membership, or generated defaults.

Release: mdstats 0.20.207a0

Dependency-graph schema: 56

Next gate: **TARGET-DATA2C-MVMIGRATE1**.